TONY GRAHAME

Experiential Learning

Students in Arizona get credit, and valuable experience, designing and building a house.

by Tony Grahame

While construction technology is advancing rapidly, the way houses are being built today is changing at a moderate pace. The design and construction of buildings has become much more complex over the past 50 years, with the introduction of insulation, forced-air systems, and tighter building envelopes, and with a more fundamental understanding of the building sciences. This complexity has created an interior microclimate that has changed the way buildings perform in their particular environments and the way occupants live in that space. To further complicate the building process, many houses have become more difficult to build, because architects are producing more complex designs (see "New Construction Report Card," *HE* Jan/Feb '03, p. 18). There is also a growing awareness of the detrimental ecological consequences of building with no consideration given to sustainable building practices.

Many of these changes in building technology have occurred in a relatively short span of time, and not all builders are keeping up with these advances. Besides, many builders say, their clients aren't asking for more energy-efficient technology, or for the use of sustainable materials. That may be true for now, but the times are changing. As our fuel resources begin to dwindle and become more expensive, builders will be facing more demand for green building, especially as more people become aware of the damage—to their environment, their health, and their bank accounts—caused by poor building practices.

In response to these concerns, a number of organizations, both private and public, have written specifications, checklists, and guidelines to help buyers determine if the house they are considering is certified as an energy-efficient or healthy house. An Energy Star-certified house is becoming synonymous with an energy-efficient house. The American Lung Association Health House organization has recently published new standards for healthy-house certification. Other organizations, too, have well-established criteria for energy-efficient and/or green-built houses.

Education Is Key

Lack of access to education is one reason why builders are not building—and home buyers are not buying—energy- and resource-efficient, environmentally responsive, healthy houses as a matter of routine. Many designers and builders have neither the inclination nor the time to go back into the classroom, but there are training

TONY GRAHAME

TONY GRAHAME

(top) Students install an entirely engineered lumber flooring system. *(below)* This 2000-ft² house was designed and built by students. Features of this house, completed in 2003, include a rainwater catchment system.

opportunities for both the novice or the experienced builder who want to build energy- and resource-efficient, healthy, and affordable housing (see "Guide to Training Programs," p. 23). One way a person can learn to build houses the right way and earn a degree while doing it is through the Construction Technology program at Yavapai College in Prescott, Arizona, where I teach.

This construction program has existed for 23 years. It is one of the most progressive and innovative building science and construction technology programs available. Each year students actually design and build a house integrating current building science principles with mainstream construction practices. Students can earn an advanced certificate or an associate of applied science degree, or prepare for transfer to a four-year university. Areas of emphasis include architectural graphics, commercial construction management, residential construction management, and residential building technology. Professionals currently working in the construction industry can enroll in specific courses and take advantage of opportunities to upgrade their existing skills or stay current with the latest advances in building technologies.

Construction Technology Program

The Construction Technology program begins when a lot is purchased in a local subdivision. Second-year architectural graphics students are then given the assignment of designing a house specifically for that lot. They are given a list of building requirements and specifications and specific technologies that must be incorporated into each student's individual design. The specs include multiple returns and a balanced air distribution system, a sealed duct system inside the conditioned space, a super insulated and airtight building envelope, radiant in-floor heating, and a heat recovery ventilation system with high-efficiency particulate air (HEPA) filtration. Add to that passive-solar design, an active-solar hot water system, high-performance windows, sealed-combustion appliances, energy-efficient lighting, no- or low-volatile organic compound (VOC) interior finishes, and rainwater catchment. The list goes on. It's a major challenge for students to incorporate these technologies into their design.

During the two semesters it takes to complete the design, faculty spend considerable time going over the requirements and technologies with the students' input. Faculty and classmates critique the designs to emulate the client/architect relationship. Students are encouraged to be creative yet meet the client's needs. At year's end, a completed house plan is selected to be constructed by residential building technology students the following semester.

Each year residential building technology students, using appropriate technologies, build a house that is energy and resource efficient, safe, comfortable, healthy, affordable, durable, and green built. Students work together to perform all the tasks required to build the house, except for the plumbing, HVAC, insulation, drywall, cabinets, stucco, and tile roof, which we subcontract. We select subcontractors who will both instruct students and allow them to assist in the installations. Students are acutely aware of air, moisture, and thermal barrier details, and therefore do the necessary prep work before the subcontractors arrive. In this way, students are encouraged to develop a basic knowledge of all the trades in order to grasp the concept that a house needs to be built as a system if it is to perform efficiently.

The houses are approximately 2,000 ft² (unless we build one with a basement, which we sometimes do) and sell for approximately $260,000. The last house (2003) cost $90/ft² to build, plus $46,000 for the lot. The profit goes

TONY GRAHAME

TONY GRAHAME

(top) Unvented, insulated stem walls and rim joists sealed the conditioned crawlspace in this house. All the ducts were located inside conditioned space. *(bottom)* Students are a part of every decision necessary to construct the house. They learn that building a house involves hundreds of important decisions.

toward scholarships. Students work at the site 20 hours a week for 15 weeks for two semesters. Because the students do so much of the work, our labor costs are low. This enables us to introduce and teach students a variety of new and appropriate technologies to which they might not otherwise be exposed.

We use the Scottsdale, Arizona Green Building program checklist and rating system to certify our houses as green built. Although we are not certified by the American Lung Association Health House organization, we use their criteria for building a healthy house, because indoor air quality is a high priority. We use the Energy and Environmental Building Association (EEBA) *Builder's Guide, Hot-Dry Climates* as one of our texts.

Report Card and Invitation

The houses are independently tested for energy efficiency through Duct Blaster and blower door tests. Advanced Insulation, Incorporated, in Prescott, performed a blower door test on our 2003 house. EEBA recommends an ACH of 3 or less, and the very stringent Canadian standard is 1.5 ACH or less. Advanced Insulation measured 1.4 ACH at a house pressure of 50 Pa, or 365 CFM_{50}. We measured the ducts in this house for leakage during the construction process, and found just over 1%, which was directly attributed to leaks in the air handler and heat recovery ventilation units. We sealed these leaks.

Advanced Insulation's final report on the 2002 house included this affirmation: "The college house is more advanced in terms of incorporation of systems and technologies to enhance the healthfulness, comfort, safety, and efficiency than the majority of the homes built in Prescott today."

Many of our graduates have gone on to become successful tradespeople, drafters, subcontractors, foremen, superintendents, general contractors, project managers, construction managers, designers, architects, building inspectors, and energy efficiency consultants. Their knowledge and enthusiasm have contributed significantly to the residential construction industry.

Newcomers and veteran builders interested in learning about building houses that are energy and resource efficient and environmentally responsive should explore the Construction Technology program at Yavapai College. Not only do we have a progressive residential construction program, but we also have a wonderful year-round climate in the mountains of north central Arizona.

Tony Grahame teaches in the Construction Technology Department at Yavapai College in Prescott, Arizona.

For more information:

Yavapai College Construction Technology Department
1100 E Sheldon St.
Prescott, AZ 86301
Tel: (928)776-2358
Fax: (928)717-7699
E-mail: tgrahame@yc.edu
Web site: www.yc.edu